GEMINI 4 ARGON SF HACKATHON · OCTOBER 25, 2026

Virtual Lab

A lab of 28 Gemini agents that plans a study, runs it on real patient data, and has to beat Gemini 4 Argon answering alone.

github.com/SankarSubbayya/virtual_lab

One research question takes a real lab days

A REAL LAB

Days

Read the papers, plan the analysis, argue about it, write the code, run it on the data. Slow, but the numbers are measured.

AN AI CHATBOT

Seconds

Answers from memory. It never opens the data, so its numbers are educated guesses.

We wanted the speed of the chatbot **and** the rigour of the lab.

So we built the lab out of AI agents

1

Study

Learns the topic and writes cited notes the team keeps

2

Plan ×3

Three meetings at once: a lead scientist and three specialists

3

Critique

A critic attacks each plan until it holds up

4

Approve

A person approves the cost before anything runs

5

Run

The code runs on 1,980 real patients

6

Prove

A blind judge compares it with Argon alone

28 agents, each with one job. They pass work along like a relay, keep shared notes, and check each other’s work.

Data: METABRIC breast cancer cohort (Curtis et al., Nature 2012), public via cBioPortal

Argon leads; the others do the legwork

Role	Model	What it does
Lead scientist, critic, coder, writer, judge	**Gemini 4 Argon**	Runs the meetings, attacks the plans, writes the analysis code, writes the answer, judges
Specialists and the school	Gemini 3.8 Flash	Ideas from immunology, statistics and data; study notes
The lab bench	**Gemini Managed Agents**	Runs the code on 1,980 patients in a secure Google sandbox
First draft	Gemma 4	A quick answer from memory, checked later
The person in charge	You	Approves the budget; the approval becomes a hard spending cap

The data: a public breast cancer cohort

2,509

patients in the METABRIC study, from tumour banks in the UK and Canada, with long follow-up

1,980

of them have tumour gene-expression data, which the immune scores need: this is the lab's dataset

320

are triple-negative (ER, PR and HER2 negative), 168 of whom died during follow-up; the lab selects them itself

Public and de-identified. Downloaded by our code from the cBioPortal API (study **brca_metabric**): clinical details, survival, and 18 immune genes per patient.

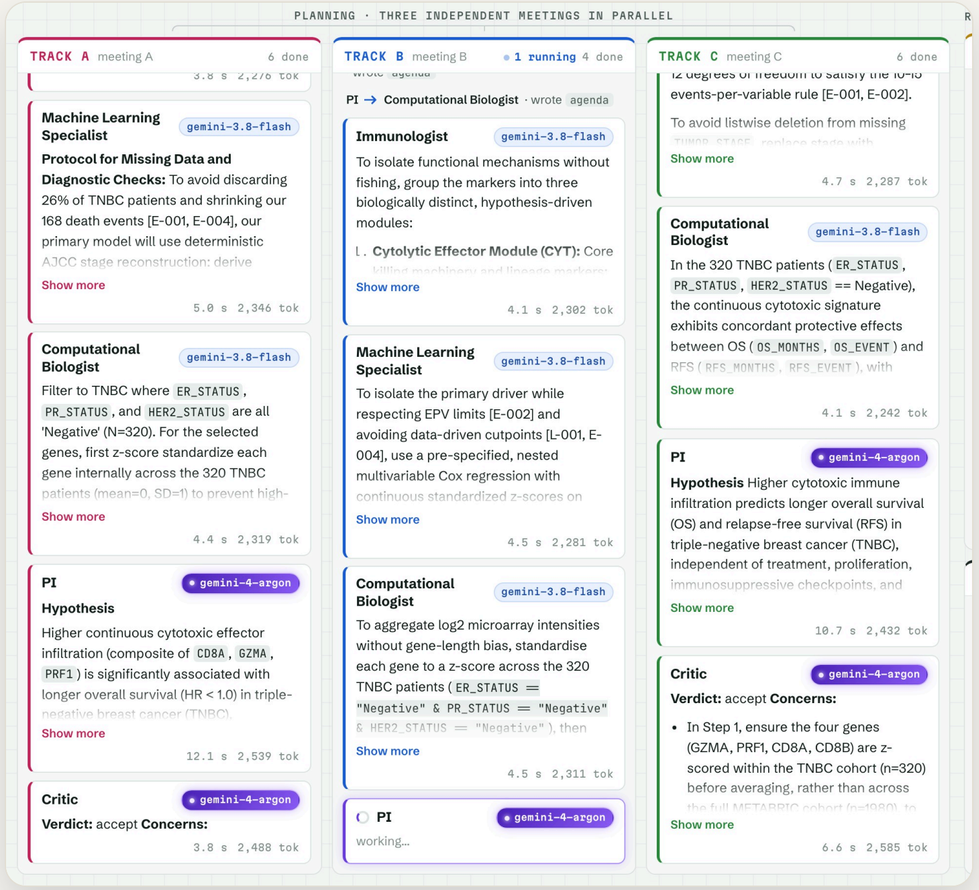
Curtis et al., Nature 2012 · Pereira et al., Nature Communications 2016 · Rueda et al., Nature 2019

LIVE DEMO · ABOUT 2.5 MINUTES

“In triple-negative breast cancer, is more immune ‘killer’ activity in the tumour linked to longer survival, after accounting for age, stage and grade?”

Watch for: three meetings at once · the approval click · code running on real data · the blind verdict

Inside a run: three meetings at once



Argon leads each meeting

A Gemini 4 Argon PI frames the problem and writes the plan.

Specialists answer in parallel

Immunologist, statistician and data expert on Gemini Flash. Each card shows its model, time and tokens.

A critic attacks every plan

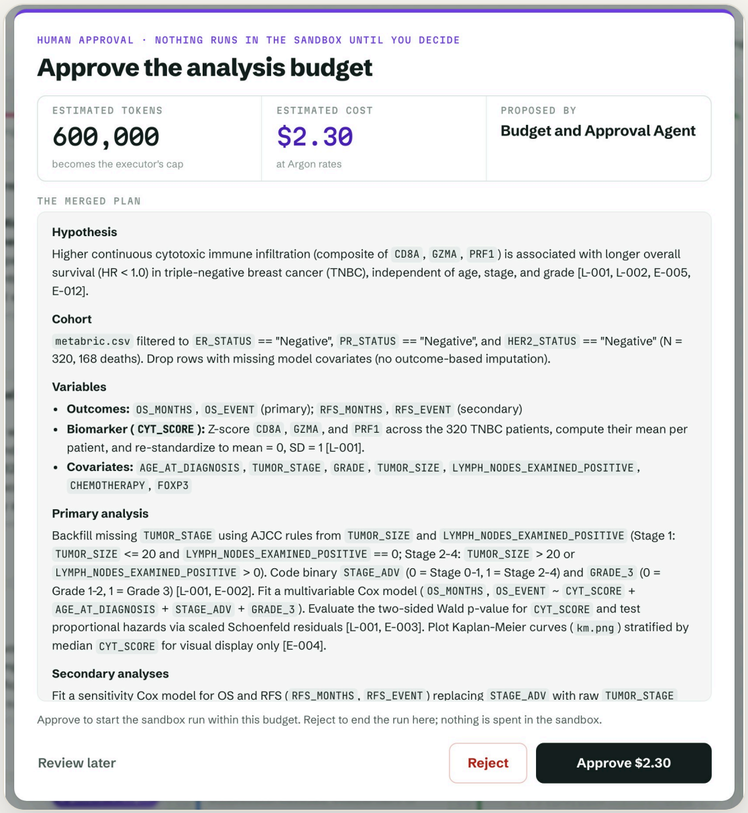
An Argon critic lists concerns, such as missing tumour stage; a reviser fixes them.

Nothing is hidden

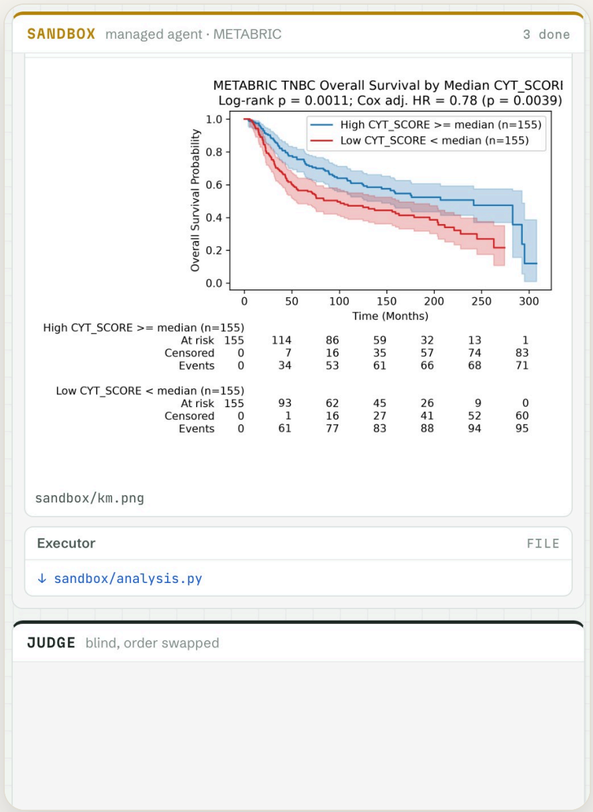
Every message and hand-off appears live and is saved to the run’s folder.

Screenshot from a live run, 45 seconds in

Approve, then run on real data



1 · The approval card: plan, cost cap, Approve or Reject



2 · The sandbox returns the survival plot and the code

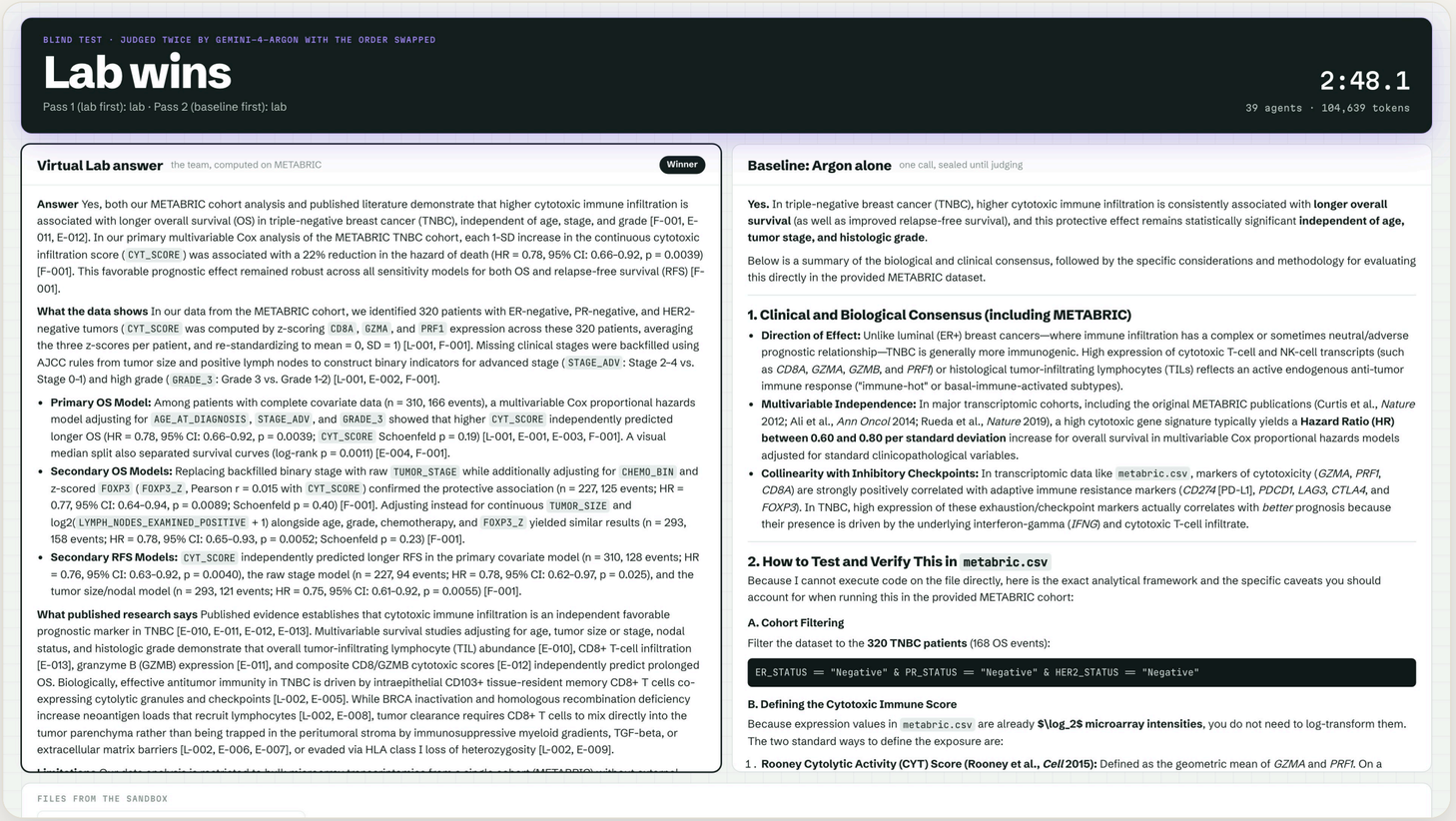
Nothing touches the data until a person approves.

The approved budget becomes a hard token cap on the Gemini managed agent.

While the card waits, Argon writes the analysis code. The agent then runs it on 1,980 patients in a Google-hosted sandbox and fixes it if it fails.

Screenshots from the same live run

The blind test: lab vs Argon alone



BLIND TEST · JUDGED TWICE BY GEMINI-4-ARGON WITH THE ORDER SWAPPED

Lab wins

Pass 1 (lab first): lab · Pass 2 (baseline first): lab

2:48.1

39 agents · 104,639 tokens

Virtual Lab answer the team, computed on METABRIC — Winner

Answer Yes, both our METABRIC cohort analysis and published literature demonstrate that higher cytotoxic immune infiltration is associated with longer overall survival (OS) in triple-negative breast cancer (TNBC), independent of age, stage, and grade [F-001, E-011, E-012]. In our primary multivariable Cox analysis of the METABRIC TNBC cohort, each 1-SD increase in the continuous cytotoxic infiltration score (`CYT_SCORE`) was associated with a 22% reduction in the hazard of death (HR = 0.78, 95% CI: 0.66–0.92, p = 0.0039) [F-001]. This favorable prognostic effect remained robust across all sensitivity models for both OS and relapse-free survival (RFS) [F-001].

What the data shows In our data from the METABRIC cohort, we identified 320 patients with ER-negative, PR-negative, and HER2-negative tumors (`CYT_SCORE` was computed by z-scoring `CD8A`, `GZMA`, and `PRF1` expression across these 320 patients, averaging the three z-scores per patient, and re-standardizing to mean = 0, SD = 1) [L-001, F-001]. Missing clinical stages were backfilled using AJCC rules from tumor size and positive lymph nodes to construct binary indicators for advanced stage (`STAGE_ADV`: Stage 2-4 vs. Stage 0-1) and high grade (`GRADE_3`: Grade 3 vs. Grade 1-2) [L-001, E-002, F-001].

- **Primary OS Model:** Among patients with complete covariate data (n = 310, 166 events), a multivariable Cox proportional hazards model adjusting for `AGE_AT_DIAGNOSIS`, `STAGE_ADV`, and `GRADE_3` showed that higher `CYT_SCORE` independently predicted longer OS (HR = 0.78, 95% CI: 0.66–0.92, p = 0.0039; `CYT_SCORE` Schoenfeld p = 0.19) [L-001, E-001, E-003, F-001]. A visual median split also separated survival curves (log-rank p = 0.0011) [E-004, F-001].
- **Secondary OS Models:** Replacing backfilled binary stage with raw `TUMOR_STAGE` while additionally adjusting for `CHEMO_BIN` and z-scored `FOXP3` (`FOXP3_Z`, Pearson r = 0.015 with `CYT_SCORE`) confirmed the protective association (n = 227, 125 events; HR = 0.77, 95% CI: 0.64-0.94, p = 0.0089; Schoenfeld p = 0.40) [F-001]. Adjusting instead for continuous `TUMOR_SIZE` and log2(`LYMPH_NODES_EXAMINED_POSITIVE` + 1) alongside age, grade, chemotherapy, and `FOXP3_Z` yielded similar results (n = 293, 158 events; HR = 0.78, 95% CI: 0.65–0.93, p = 0.0052; Schoenfeld p = 0.23) [F-001].
- **Secondary RFS Models:** `CYT_SCORE` independently predicted longer RFS in the primary covariate model (n = 310, 128 events; HR = 0.76, 95% CI: 0.63–0.92, p = 0.0040), the raw stage model (n = 227, 94 events; HR = 0.78, 95% CI: 0.62–0.97, p = 0.025), and the tumor size/nodal model (n = 293, 121 events; HR = 0.75, 95% CI: 0.61–0.92, p = 0.0055) [F-001].

What published research says Published evidence establishes that cytotoxic immune infiltration is an independent favorable prognostic marker in TNBC [E-010, E-011, E-012, E-013]. Multivariable survival studies adjusting for age, tumor size or stage, nodal status, and histologic grade demonstrate that overall tumor-infiltrating lymphocyte (TIL) abundance [E-010], CD8+ T-cell infiltration [E-013], granzyme B (GZMB) expression [E-011], and composite CD8/GZMB cytotoxic scores [E-012] independently predict prolonged OS. Biologically, effective antitumor immunity in TNBC is driven by intraepithelial CD103+ tissue-resident memory CD8+ T cells co-expressing cytolytic granules and checkpoints [L-002, E-005]. While BRCA inactivation and homologous recombination deficiency increase neoantigen loads that recruit lymphocytes [L-002, E-008], tumor clearance requires CD8+ T cells to mix directly into the tumor parenchyma rather than being trapped in the peritumoral stroma by immunosuppressive myeloid gradients, TGF-beta, or extracellular matrix barriers [L-002, E-006, E-007], or evaded via HLA class I loss of heterozygosity [L-002, E-009].

Baseline: Argon alone one call, sealed until judging

Yes. In triple-negative breast cancer (TNBC), higher cytotoxic immune infiltration is consistently associated with **longer overall survival** (as well as improved relapse-free survival), and this protective effect remains statistically significant **independent of age, tumor stage, and histologic grade**.

Below is a summary of the biological and clinical consensus, followed by the specific considerations and methodology for evaluating this directly in the provided METABRIC dataset.

1. Clinical and Biological Consensus (including METABRIC)

- **Direction of Effect:** Unlike luminal (ER+) breast cancers—where immune infiltration has a complex or sometimes neutral/adverse prognostic relationship—TNBC is generally more immunogenic. High expression of cytotoxic T-cell and NK-cell transcripts (such as *CD8A*, *GZMA*, *GZMB*, and *PRF1*) or histological tumor-infiltrating lymphocytes (TILs) reflects an active endogenous anti-tumor immune response ("immune-hot" or basal-immune-activated subtypes).
- **Multivariable Independence:** In major transcriptomic cohorts, including the original METABRIC publications (Curtis et al., *Nature* 2012; Ali et al., *Ann Oncol* 2014; Rueda et al., *Nature* 2019), a high cytotoxic gene signature typically yields a **Hazard Ratio (HR) between 0.60 and 0.80 per standard deviation** increase for overall survival in multivariable Cox proportional hazards models adjusted for standard clinicopathological variables.
- **Collinearity with Inhibitory Checkpoints:** In transcriptomic data like `metabric.csv`, markers of cytotoxicity (*GZMA*, *PRF1*, *CD8A*) are strongly positively correlated with adaptive immune resistance markers (*CD274* [PD-L1], *PDCD1*, *LAG3*, *CTLA4*, and *FOXP3*). In TNBC, high expression of these exhaustion/checkpoint markers actually correlates with *better* prognosis because their presence is driven by the underlying interferon-gamma (*IFNG*) and cytotoxic T-cell infiltrate.

2. How to Test and Verify This in `metabric.csv`

Because I cannot execute code on the file directly, here is the exact analytical framework and the specific caveats you should account for when running this in the provided METABRIC cohort:

A. Cohort Filtering

Filter the dataset to the **320 TNBC patients** (168 OS events):

```
ER_STATUS == "Negative" & PR_STATUS == "Negative" & HER2_STATUS == "Negative"
```

B. Defining the Cytotoxic Immune Score

Because expression values in `metabric.csv` are already **$\log_2$ microarray intensities**, you do not need to log-transform them. The two standard ways to define the exposure are:

1. **Rooney Cytolytic Activity (CYT) Score (Rooney et al., *Cell* 2015):** Defined as the geometric mean of *GZMA* and *PRF1*. On a

FILES FROM THE SANDBOX

JUDGED TWICE, ORDER SWAPPED

Lab 24, Argon 16
Lab 24, Argon 19

"B only describes how the analysis should be conducted without running the data, and even makes an empirical claim about FOXP3 collinearity that A's real data analysis disproves."

The blind judge (Gemini 4 Argon) on this run

Screenshot from the same live run, 2 minutes 48 seconds in

The lab wins because it measures

10 of 11

blind tests won against Argon alone, judged twice with the order swapped (the lab must win both)

~20% lower

risk of death with more immune ‘killer’ activity: hazard ratio 0.76–0.80 in every run, measured on 227–320 real patients

2.5 min

from question to verdict, including the human approval (it took 11 minutes this morning)

$0.35

per question, about 120,000 tokens

Data: METABRIC breast cancer cohort, public via cBioPortal

Safe and fair by design

No medical advice

A question about one patient is turned into a research question about patients in general. Public data only.

A person holds the purse

Nothing runs on the data until someone approves the cost, and the approved budget is a hard cap.

A fair fight

The rival is Argon itself, with the same question and data description. The judge is blind, sees both orders, and the lab must win both.

What we built today

ALL OF THE CODE, WRITTEN TODAY

- The 28-agent relay on the Gemini Interactions API
- Analysis on Gemini Managed Agents, capped by the approved budget
- Fast mode: from 11 minutes to 2.5
- A live page showing every hand-off
- Safety guard, blind judge, 27 tests, two independent code reviews

github.com/SankarSubbayya/virtual_lab

BEFORE TODAY

An earlier prototype of the idea (Gemma and Claude). It never ran code on data and lost its blind test. We kept the design and rewrote everything.